

Financial Frictions, Used Capital, and Misallocation

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Job Market Paper

Version: September 18, 2026

Abstract. Standard macrofinance models treat capital as homogeneous. Financially constrained firms can therefore adjust only the quantity of capital they operate, not its vintage. This assumption is at odds with the data: young, small, and credit-excluded firms use second-hand equipment disproportionately. I build a dynamic macro-finance model in which firms rent both new and used capital. Second-hand capital lets constrained firms operate closer to their efficient scale, even when they cannot meet the collateral requirements for new capital. It thereby reduces the dispersion of marginal revenue products of capital and weakens the dependence of entrepreneurship on initial wealth. I calibrate the model to Vietnamese firms and find that used capital mitigates the allocative and occupational distortions created by financial frictions.

JEL codes: E22, E44, L26, O47.

Keywords: Financial frictions, used capital, misallocation, entrepreneurship, aggregate productivity.

*I am deeply grateful to my supervisors, Tasso Adamopoulos and Chaoran Chen, for their guidance and support, and to my committee member, Yanran Guo, for her helpful comments and suggestions. I also thank Jeremy Lin for his encouragement and support. I thank the MDG group, participants in the macroeconomics seminars at York University, and participants at the Canadian Economics Association conference for their valuable feedback. All remaining errors are my own.

Capital is misallocated across firms in developing countries, reducing aggregate total factor productivity. A large literature identifies capital misallocation as an important source of these productivity losses (David and Venkateswaran, 2019; Bau and Matray, 2023). Financial frictions are a natural source of such misallocation, and a large quantitative literature studies them in dynamic general equilibrium models with collateral constraints (Buera et al., 2011; Moll, 2014; Midrigan and Xu, 2014). A common feature of these models is that capital is homogeneous: a financially constrained firm can adjust the quantity of capital it operates, but not whether that capital is new or used. This omits an adjustment margin that matters in the data, since young and financially constrained firms rely disproportionately on used equipment (Eisfeldt and Rampini, 2007; Ma et al., 2022). Whether this margin changes the aggregate consequences of financial frictions is an open quantitative question.

I address this question by introducing a choice between new and used capital into a dynamic general equilibrium model with collateral constraints and endogenous occupational choice. Firms rent both new and used capital from a financial intermediary, choosing the mix jointly with their operating scale. The model therefore links firms' use of used equipment to their productivity and financial position. It also allows a second-hand capital to affect both the allocation of capital across operating firms and the decision to become an entrepreneur.

The central mechanism is a trade-off between the rental cost of used equipment and the collateral required to rent it. Used equipment may carry a higher rental cost per physical unit, but because its market value is lower it absorbs less of a constrained entrepreneur's financing capacity per dollar of rental spending. A productive entrepreneur with little wealth may consequently rent used capital and operate at a scale that new capital alone would not finance. As wealth rises, the collateral constraint becomes less important and firms' choices converge toward the unconstrained mix. The same mechanism raises the value of entrepreneurship for productive but poor agents and thus weakens the dependence of occupational choice on initial wealth.

I quantify this mechanism using firm-level data on Vietnamese small and medium-sized enterprises. The benchmark economy contains both new and used capital; the comparison economy removes a second-hand capital. I calibrate the two separately, so that each reproduces the same five targets, including the debt-to-output ratio and the top-decile employment share; the benchmark additionally matches the decline in used-capital reliance with firm assets and the aggregate ratio of new to used capital. I then remove the collateral constraint within each calibrated economy and recompute equilibrium prices, occupational choices, and stationary distributions. The comparison identifies both the contribution of a second-hand capital and the cost of the collateral

constraint.

Quantitatively, used capital has two effects. First, it improves how capital is allocated: relative to a single-vintage economy – one in which firms rent only new capital, calibrated on its own to the same five empirical targets – a second-hand capital reduces the misallocation that collateral constraints generate by 39 percent. Second, it changes who runs a firm. Collateral constraints keep many talented but poor agents out of entrepreneurship, and a second-hand capital repairs 23 percent of that lost entry, while entry by talented rich agents is nearly unchanged. Entrepreneurship therefore depends much less on initial wealth.

This paper contributes to the literature on factor misallocation and financial frictions, in which a constrained firm can adjust only the quantity of capital it operates.² This paper adds the choice of which kind of capital to operate, and that choice turns out to matter: access to second-hand capital removes a substantial part of the misallocation that collateral constraints generate in an economy without a second-hand capital. The paper also relates to research on entrepreneurship and wealth.³ I differ from that work in giving the constrained agent a second capital type rather than only a smaller scale, and I show that this changes who becomes an entrepreneur: because used equipment absorbs less financing capacity per dollar of rental spending, a talented agent can enter on less wealth than the collateral constraint would otherwise require.

Section I presents the motivating evidence, Section II develops the model, Section III presents the quantitative analysis, and Section IV concludes.

I. Evidence on Used Capital and Financing Frictions

I use the Vietnamese Small and Medium Enterprise Survey (CIEM et al., 2005–2015). Table 1 reports the used-capital share by firm age. Second-hand equipment accounts for 31.5 percent of a typical young firm’s main equipment, against 21.6 percent for firms past eleven years. Similar gradients appear in richer economies (Eisfeldt and Rampini, 2007; Ma et al., 2022).

The financing mechanism predicts that firms without debt rely more on cheaper used machines when their assets are low, and less so as collateral accumulates. I

²On the measurement of misallocation and its aggregate cost, see Restuccia and Rogerson (2008), Hsieh and Klenow (2009, 2014), David and Venkateswaran (2019), Bau and Matray (2023), and Adamopoulos and Restuccia (2014). On financial frictions in dynamic general equilibrium models with collateral constraints, see Buera et al. (2011), Moll (2014), Midrigan and Xu (2014), and Chen et al. (2023); Buera et al. (2015) survey this macrodevelopment literature.

³Evans and Jovanovic (1989), Cagetti and De Nardi (2006), Buera and Shin (2013), and Tan (2025).

Table 1. Used-capital share by firm age

Firm age	Share (%)
Young (0–5 years)	31.5
6–10 years	26.0
11 years or more	21.6

Notes: The table reports the average used-capital share by firm-age group. Author’s calculations based on the pooled 2005–2015 waves of the Vietnam SME Survey (CIEM et al., 2005–2015).

estimate

$$\text{UsedShare}_{it} = \beta_1 \text{ZeroDebt}_{it} + \beta_2 (\text{ZeroDebt}_{it} \times X_{it}) + \beta_3 X_{it} + \mathbf{\Gamma}' \mathbf{Z}_{it} + \lambda_t + \alpha_i + \varepsilon_{it}, \quad (1)$$

where UsedShare_{it} is used capital as a percentage of firm i ’s main equipment or machinery, $\mathbf{Z}_{it}$ holds two employment-size indicators, and the columns differ only in X_{it} : centered log assets in column (1), and an indicator for firms aged five years or less in column (2). Both columns include firm and year fixed effects, so identification comes from variation within firms over time, and standard errors are clustered by firm.

Table 2. Financing frictions and used capital

<i>Dependent variable:</i> used-capital share (percentage points)		
	(1) Asset interaction	(2) Young-firm interaction
Zero debt $\times$ log assets	−1.067** (0.462)	—
Zero debt $\times$ young	—	3.885* (2.340)
Zero debt	−0.482 (0.836)	−1.046 (0.872)
Log assets	−0.260 (0.493)	−0.679 (0.453)
Young (age ≤ 5)	—	0.755 (1.658)
Observations	13,492	13,403
Firms	3,551	3,549
Within R^2	0.036	0.036
Firm fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Size controls	Yes	Yes

Notes: Standard errors clustered by firm are in parentheses. Sector and province are absorbed by the firm fixed effects. Log assets is the firm’s reported total assets, centered at the sample mean. The size controls are the two employment-size indicators in $\mathbf{Z}_{it}$: one for firms with 10–49 workers and one for firms with 50 or more, micro firms being the omitted category; neither is individually significant in either column and their coefficients are not tabulated. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The estimates support the collateral prediction on both margins. The extra used capital a firm holds while carrying no debt shrinks by 1.067 percentage points for each log point of assets it accumulates (column 1), and is 3.885 points larger while the firm is young (column 2). Both are measured within firms, so they compare a firm with itself as it grows and ages rather than small firms with large ones. The evidence is descriptive rather than causal, as zero debt is an imperfect proxy for credit access.⁴

II. Model

I develop a stationary general equilibrium model with endogenous occupational choice and a competitive rental market for new and used capital. Agents save through deposits with a competitive financial intermediary that owns the physical capital stock and rents capital services to entrepreneurs. All agents, including the intermediary, are price takers, and all rental rates and asset prices are equilibrium objects. The model departs from a standard collateral-constraint economy in one respect: firms rent both new and used capital, and the collateral constraint is written on the combined market value of the two. Throughout, capital of type $j \in \{n, u\}$ has price p_j and user cost R_j per dollar of its market value, so the rental payment per physical unit is $p_j R_j$.

A. Environment and Timing

Time is discrete and there is no aggregate uncertainty. A unit measure of infinitely lived agents is indexed by the individual state (a, z, y) , where a is financial wealth, z is log entrepreneurial productivity, and y is worker ability. Preferences are

$$U_0 = E_0 \sum_{t=0}^{\infty} \beta^t u(c_t), \quad u(c) = \frac{c^{1-\sigma} - 1}{1 - \sigma}, \quad (2)$$

with discount factor $\beta \in (0, 1)$ and risk aversion σ .

B. Technology and Idiosyncratic States

Entrepreneurial technology. An entrepreneur with productivity z hires L efficiency units of labor and rents new and used capital, K_n and K_u , to produce

$$Y = \exp(z) (L^\alpha K^{1-\alpha})^\eta, \quad \eta < 1, \quad (3)$$

where α is the labor share within the variable-input bundle and $\eta < 1$ governs the span of control, as in [Lucas \(1978\)](#), so that optimal firm scale is finite and increasing

⁴A more direct measure gives the same answer. Among formal-loan applicants in 2013, firms reporting difficulty obtaining a loan hold 7.9 percentage points more used capital than observationally similar firms ($N = 641$); the pooled 2011, 2013 and 2015 estimate is 3.7 points.

in productivity. New and used capital are imperfect substitutes, with effective capital given by the constant-elasticity aggregate

$$K = \left[\phi^{1/\gamma} K_n^{(\gamma-1)/\gamma} + (1-\phi)^{1/\gamma} K_u^{(\gamma-1)/\gamma} \right]^{\gamma/(\gamma-1)}, \quad (4)$$

where $\gamma > 1$ is the elasticity of substitution between new and used capital and ϕ their relative weight. Entrepreneurial productivity is persistent:

$$z' = \begin{cases} z, & \text{with probability } \omega, \\ \tilde{z}', \quad \tilde{z}' \sim F_z, & \text{with probability } 1 - \omega, \end{cases} \quad (5)$$

so ω governs the persistence of productivity. The induced distribution of productivity levels, $\exp(z)$, has a Pareto right tail with parameter η_p .

Workers. A worker with ability y supplies y efficiency units inelastically and earns

$$D_w(y) = Wy, \quad (6)$$

where W is the wage per efficiency unit. Ability follows

$$\log y' = \rho_y \log y + \sigma_y \varepsilon'_y, \quad \varepsilon'_y \sim N(0, 1), \quad (7)$$

with stationary standard deviation $\sigma_y / \sqrt{1 - \rho_y^2}$. Productivity and ability evolve independently.

C. Individual Decisions and Occupational Choice

Entrepreneurial profits. Conditional on operating a firm, an agent with wealth a and productivity z chooses labor and both new and used capital to maximize operating profits,

$$\pi(a, z) = \max_{K_n, K_u, L \geq 0} \left\{ \exp(z) \left(L^\alpha K^{1-\alpha} \right)^\eta - WL - p_n R_n K_n - p_u R_u K_u \right\}, \quad (8)$$

subject to K in (4) and the financing constraint

$$p_n K_n + p_u K_u \leq \theta a, \quad (9)$$

where a is the entrepreneur's asset holdings used as collateral and $\theta \geq 1$ governs the stringency of the collateral constraint, with a smaller θ indicating a tighter constraint. New and used capital each absorb financing capacity equal to their own market value.

Consumption and saving. The value conditional on entrepreneurship is

$$V_e(a, z, y) = \max_{\substack{c \geq 0 \\ a' \geq 0}} \{u(c) + \beta E[\Omega(a', z', y') \mid z, y]\}, \quad (10)$$

subject to

$$c + a' = \pi(a, z) + (1 + r)a, \quad (11)$$

and the value conditional on wage work is

$$V_w(a, z, y) = \max_{\substack{c \geq 0 \\ a' \geq 0}} \{u(c) + \beta E[\Omega(a', z', y') \mid z, y]\}, \quad (12)$$

subject to

$$c + a' = Wy + (1 + r)a. \quad (13)$$

Deposits are non-negative: agents cannot borrow against future labor income, and the only credit in the economy is the intermediated capital limited by (9).

Occupational choice. The value available at (a, z, y) is

$$\Omega(a, z, y) = \max \{V_e(a, z, y), V_w(a, z, y)\}, \quad (14)$$

and the agent operates a firm if

$$V_e(a, z, y) \geq V_w(a, z, y) \quad (15)$$

and supplies labor otherwise. These occupational flows, together with the resulting composition of the entrepreneurial sector, are equilibrium outcomes of (15).

D. The Financial Intermediary and Capital Prices

A competitive financial intermediary collects deposits at the risk-free rate r , owns the physical capital stock, and rents capital services to entrepreneurs. It earns zero profit. The final good is the numeraire and one unit of it becomes one unit of new capital, so $p_n = 1$ throughout. The stocks evolve as

$$K_{n,t+1}^s = (1 - \delta - \zeta)K_{n,t}^s + I_t, \quad (16)$$

$$K_{u,t+1}^s = (1 - \delta - \kappa)K_{u,t}^s + \zeta K_{n,t}^s, \quad (17)$$

where I_t is investment, δ is baseline depreciation, $\zeta \in (0, 1)$ is the fraction of new capital that becomes used each period, and κ is the fraction of used capital that breaks

and is hence valueless, with $\delta + \zeta < 1$ and $\delta + \kappa < 1$. Used capital is formed from new capital that becomes used, so it is in positive supply without absorbing investment of its own. Write p_u for the price of used capital relative to new, so that $p_u R_u$ is the rental payment per physical unit.

Proposition 1 (Capital prices). *Let the intermediary discount at r , take prices as given, and choose investment to maximize the present value of rental income net of investment subject to (16)–(17), where new capital rents at $p_n R_n$ and used capital at $p_u R_u$ per physical unit. Then a stationary solution with $K_n^s, K_u^s \in (0, \infty)$ requires*

$$R_n = r + \delta + \zeta \left(1 - \frac{p_u R_u}{r + \delta + \kappa} \right). \quad (18)$$

The proof is in Appendix A.

Only the product $p_u R_u$ appears in (18), and only that product enters a firm's rental bill, so the model determines the rental payment on used capital rather than its two factors. Define

$$R_u = r + \delta + \kappa, \quad (19)$$

the user cost of used capital: one dollar held in used capital must earn the deposit rate, cover depreciation, and cover the rate at which the machine breaks. With R_u so defined, (18) determines the price of used capital relative to new,

$$p_u = p_n - \frac{R_n - r - \delta}{\zeta}, \quad (20)$$

and (18) can be written $R_n = r + \delta + \zeta(p_n - p_u)$. The last term is the net cost of aging: with probability ζ a unit priced at p_n is repriced at p_u . At $\zeta = 0$ it collapses to $R_n = r + \delta$, which is the case in the economy without used capital.

This split would be immaterial if used capital merely produced. It matters here because the financing constraint (9) is written on market value, which makes p_u enter the entrepreneur's problem on its own. The mechanism of the paper originates in it: breakage raises R_u through (19), making used capital expensive to rent, while the price at which its rental stream capitalizes is determined separately by (20), making it cheap in collateral capacity.

Deposits are the intermediary's only liability and capital its only asset, so

$$D = p_n K_n^s + p_u K_u^s. \quad (21)$$

Given (21), the pricing condition implies zero profit, as Appendix A shows. The intermediary is a veil: the economy behaves as if households held the capital stock

directly and priced it at (p_n, p_u) .

In a stationary equilibrium the used stock is constant, so

$$\zeta K_n^s = (\delta + \kappa) K_u^s, \quad \text{hence} \quad \frac{K_u^s}{K_n^s} = \frac{\zeta}{\delta + \kappa}. \quad (22)$$

This ratio is fixed by (ζ, δ, κ) and does not respond to prices; the levels of both stocks scale with aggregate wealth, as (27) makes explicit. Three prices clear three markets: the wage clears the labor market, the deposit rate clears the new-capital market through (21), and R_n adjusts until the demand ratio matches (22). By (20) a higher R_n lowers p_u , which tilts demand toward used capital through both the rental payment and the collateral a unit absorbs, so used-capital clearing is automatic.

E. Aggregation and Stationary Equilibrium

Let $G(a, z, y)$ denote the stationary distribution of agents. Occupation is a function of the current state and so is not a separate coordinate of G . Define

$$\mathcal{E} = \{(a, z, y) : V_e(a, z, y) \geq V_w(a, z, y)\}, \quad (23)$$

$$\mathcal{W} = \{(a, z, y) : V_e(a, z, y) < V_w(a, z, y)\}. \quad (24)$$

Aggregate input demands and labor supply are

$$K_n^d = \int_{\mathcal{E}} K_n(a, z) dG, \quad K_u^d = \int_{\mathcal{E}} K_u(a, z) dG, \quad (25)$$

$$L^d = \int_{\mathcal{E}} L(a, z) dG, \quad L^s = \int_{\mathcal{W}} y dG. \quad (26)$$

Total deposits are $D = \int a dG$. By the balance-sheet condition (21), deposits equal the market value of the capital the intermediary holds. Combining this with the supply ratio (22) gives both stocks as functions of aggregate wealth:

$$K_n^s = \frac{D}{p_n + p_u \zeta / (\delta + \kappa)}, \quad K_u^s = \frac{\zeta}{\delta + \kappa} K_n^s. \quad (27)$$

Aggregate external finance is the value of operated capital in excess of own net worth,

$$D^{\text{ext}} = \int_{\mathcal{E}} \max \{0, p_n K_n(a, z) + p_u K_u(a, z) - a\} dG, \quad (28)$$

and its ratio to output is the leverage moment that disciplines θ . Markets clear when

$$K_n^d = K_n^s, \quad K_u^d = K_u^s, \quad L^d = L^s. \quad (29)$$

Given (27), these three conditions are equivalent to clearing the labor market, clearing the new-capital, and requiring aggregate demand to reproduce the technological ratio $K_n^d/K_u^d = (\delta + \kappa)/\zeta$. Used-capital clearing is then automatic.

Definition. A stationary competitive equilibrium consists of prices (W, r, R_n) , the implied capital prices $p_u = p_n - (R_n - r - \delta)/\zeta$ and $R_u = r + \delta + \kappa$, value functions $\{V_e, V_w, \Omega\}$, policy functions, firm input demands, capital stocks (K_n^s, K_u^s) , and a distribution G such that:

1. given (W, R_n, R_u, p_u) , firm input demands solve (8)–(9);
2. given prices and the continuation value Ω , consumption, saving, and occupational policies solve (10)–(15);
3. the intermediary’s investment policy is optimal given (r, R_n) , which by Proposition 1 requires (19)–(18) and delivers zero profit and the balance sheet (21), with the stocks evolving according to (16)–(17);
4. G is invariant under the optimal saving and occupational policies and the transition processes (5) and (7);
5. the labor, new-capital, and used-capital clear as in (29).

F. Used Capital and Financial Constraints

The capital mix. Let $\mu(a, z) \geq 0$ be the multiplier on (9). For an interior choice of both new and used capital, the firm’s first-order conditions are

$$\text{MPK}_n = p_n(R_n + \mu), \quad \text{MPK}_u = p_u(R_u + \mu). \quad (30)$$

Both capital types are used in positive amounts at any finite shadow prices. Although $\gamma > 1$ makes the two types gross substitutes, the marginal product of each remains unbounded at zero: as $K_n \rightarrow 0$ with $K_u > 0$ fixed, effective capital converges to $(1 - \phi)^{1/(\gamma-1)}K_u > 0$, so $\partial K/\partial K_n = \phi^{1/\gamma}(K/K_n)^{1/\gamma} \rightarrow \infty$, and symmetrically for K_u . Both conditions in (30) therefore hold with equality at any finite shadow prices. The two capital types are treated symmetrically. The firm pays $p_j R_j$ to rent one unit of type j and bears the additional shadow cost $p_j \mu$ for the financing capacity that the unit absorbs. Neither type receives a preferential collateral coefficient; they differ only in the market value p_j at which they enter (9). Combining (30) with (4) gives the optimal capital mix $\rho \equiv K_u/K_n$:

$$\rho(\mu) = \frac{1 - \phi}{\phi} \left[\frac{p_n(R_n + \mu)}{p_u(R_u + \mu)} \right]^\gamma. \quad (31)$$

At $\mu = 0$, this becomes $\rho^{uc} \equiv \rho(0)$. Because it depends only on market prices, all unconstrained firms choose the same mix of new and used capital, regardless of their wealth or productivity. The mix varies across firms only through the multiplier μ .

Collateral productivity. How a firm splits its capital between new and used machines also determines the scale that a given amount of financing capacity can support. Define

$$\Lambda(\rho) \equiv \frac{K}{p_n K_n + p_u K_u} = \frac{\Psi(\rho)}{p_n + p_u \rho}, \quad \Psi(\rho) \equiv \left[\phi^{1/\gamma} + (1 - \phi)^{1/\gamma} \rho^{(\gamma-1)/\gamma} \right]^{\gamma/(\gamma-1)}. \quad (32)$$

The function $\Lambda(\rho)$ measures the units of effective capital obtained per dollar of financing capacity. Define $\Lambda^{uc} \equiv \Lambda(\rho^{uc})$. A firm with a binding constraint exhausts (9) and therefore operates $\Lambda(\rho(\mu))\theta a$ units of effective capital. Appendix B derives the corresponding policies in closed form conditional on the multiplier, which is itself the root of a scalar equation. A constrained firm therefore has two adjustment margins: its scale and its choice of new versus used capital, which determines how far a given collateral position stretches.

Proposition 2 (Used-capital tilt and collateral relief). *Suppose*

$$R_u > R_n, \quad (33)$$

or, equivalently, $\kappa > \zeta(p_n - p_u)$. A sufficient condition is $\zeta \leq \kappa$, which by the stationary aggregate-stock relation (22) can equivalently be written as

$$\frac{K_u^s}{K_n^s} \leq \frac{\kappa}{\delta + \kappa}.$$

Then:

- (i) $\rho'(\mu) > 0$: *the used-to-new capital ratio increases strictly and monotonically from ρ^{uc} toward*

$$\bar{\rho} = \frac{1 - \phi}{\phi} \left(\frac{p_n}{p_u} \right)^\gamma,$$

the mix that minimizes the collateral value $p_n K_n + p_u K_u$ required per unit of effective capital.

- (ii) *Collateral productivity $\Lambda(\rho(\mu))$ is strictly increasing in μ on $[0, \infty)$ and converges to*

$$\bar{\Lambda} = \left[\phi p_n^{1-\gamma} + (1 - \phi) p_u^{1-\gamma} \right]^{1/(\gamma-1)}. \quad (34)$$

Moreover, $\bar{\Lambda} > p_n^{-1}$ whenever $p_u < p_n$ and $\gamma > 1$.

The proof is in Appendix B. Proposition 2 implies that constrained firms choose a larger used share than otherwise identical unconstrained firms, and that this share rises with the tightness of the constraint. This is the model counterpart of the firm-level gradients in Section I.⁵ The tilt does not require used capital to be cheaper to rent. Condition (33) is compatible with $p_u R_u > p_n R_n$: what matters is the financing capacity a unit absorbs per dollar of rental spending, not the rental payment itself.⁶ The relief is nevertheless bounded: because $\Lambda(\rho(\mu))$ converges to the finite limit in (34), tilting toward used capital economizes on financing capacity but cannot fully substitute for scale.⁷

Two margins. Used capital therefore affects both firm scale and occupational selection. Along the intensive margin, it allows constrained entrepreneurs to operate closer to their unconstrained scale, reducing the dispersion of marginal revenue products. Along the extensive margin, it raises the value of entrepreneurship for productive but poor potential entrants and shifts the occupational boundary (15) accordingly.

III. Quantitative Analysis

Assigned parameters. Seven parameters are assigned outside the calibration. Following the literature, I set the labor share $\alpha = 2/3$, annual depreciation $\delta = 0.06$, the span of control $\eta = 0.79$, and relative risk aversion $\sigma = 1.5$, all adopted from Buera et al. (2011), and the annual persistence of worker ability $\rho_y = 0.95$, from Cagetti and De Nardi (2006). The remaining two govern the new-and-used capital aggregate (4). I set the CES weight on new capital to $\phi = 0.70$. This weight is pinned rather than estimated, because it is only weakly identified. The aggregate new-to-used ratio is a market-clearing condition that ζ delivers at any value of ϕ , so ϕ reaches the targeted moments only through the used-share gradient, and it moves that moment slowly: Appendix D raises ϕ from 0.70 to 0.75, the weight used by Lanteri (2018) for a second-hand capital, leaving the headline results unchanged. I set the elasticity of substitution to $\gamma = 2$. Edgerton (2011) estimates this elasticity from relative-price regressions derived from a CES composite of new and used equipment, and reports values between 1.7 and 2.0 for farm machinery and between 1.9 and 2.4

⁵Condition (33) holds at the parameters calibrated in Section III.

⁶Eisfeldt and Rampini (2007) also predict that constrained firms use more used capital, through the timing of cash flows relative to purchase cost rather than through collateral capacity.

⁷The mechanism belongs to the market-value collateral framework of Rampini and Viswanathan (2013) and is closest to Rampini (2019), where an asset's durability determines the down payment it requires. Here durability varies *within* a single asset class that firms combine, so the response is a change in the mix rather than a move between asset classes.

for construction machinery; $\gamma = 2$ lies in the overlap.

Jointly calibrated parameters. Seven parameters are calibrated jointly in the benchmark: $\{\theta, \eta_p, \omega, \sigma_y, \beta, \kappa, \zeta\}$. Each is disciplined primarily by one moment, though all seven are selected together. The debt-to-output ratio of 0.600 disciplines the collateral multiplier θ . The employment share of the largest decile of firms, 0.632, disciplines the Pareto-tail parameter η_p of entrepreneurial productivity. The annual firm exit rate of 0.098, which I compute from the survey and which is consistent with Islamaj et al. (2024), disciplines productivity persistence ω . Agents in the model are infinitely lived and firms do not die, so the moment is mapped to the rate at which entrepreneurs switch to wage work; the survey records the closure or suspension of the household’s enterprise rather than the disappearance of the household, which is what that switch is here. The worker wage Gini of 0.372 (Doan et al., 2018) disciplines the innovation dispersion σ_y of worker ability. The annual real interest rate of 0.031, the Vietnamese deposit rate net of GDP-deflator inflation (World Bank, 2026), is matched by the discount factor: the model requires $\beta = 0.858$ to clear the deposit market at that rate.

Used-capital parameters. Two moments jointly determine κ and ζ . The first is the aggregate ratio of new to used capital, which in a stationary equilibrium is

$$\frac{K_n^s}{K_u^s} = \frac{\delta + \kappa}{\zeta}, \quad (35)$$

and captures the relative *quantity* of the two capital types in the economy. The second captures the distribution of new and used capital across firms using the cross-sectional relationship between the used-capital share and firm size:

$$s_{u,i} = b_0 + b_1 \log A_i + \varepsilon_i, \quad b_1 = -0.027 \text{ (0.0061)}, \quad N = 886. \quad (36)$$

I estimate this relationship by ordinary least squares using the 2013 cross-section of firms reporting a positive used-capital share. Whereas the first moment captures the relative quantity of new and used capital, the second primarily reflects their relative price in collateral terms. If used capital is cheaper, smaller firms, which are more likely to face binding collateral constraints, rely more heavily on it, making b_1 more negative. Conversely, a smaller price advantage for used capital produces a flatter size gradient. The two moments imply $\kappa = 0.143$ and $\zeta = 0.079$. At these values, the condition of Proposition 2 is satisfied: $R_u = r + \delta + \kappa = 0.235$ exceeds $R_n = r + \delta + \zeta(p_n - p_u) = 0.119$, and the sufficient condition $\zeta \leq \kappa$ holds as well.

Table 3. Calibration: Parameters, Values, and Targets

Parameter	Benchmark	Model without used capital	Source or target
<i>Panel A: Parameters assigned from the literature</i>			
α : Labor share	0.667	0.667	Buera et al. (2011)
δ : Depreciation rate	0.060	0.060	Buera et al. (2011)
η : Span of control	0.790	0.790	Buera et al. (2011)
σ : Risk aversion	1.500	1.500	Buera et al. (2011)
ρ_y : Ability persistence	0.950	0.950	Cagetti and De Nardi (2006)
ϕ : CES weight on new capital	0.700	1.000	Assigned; see text
γ : Substitution, new vs. used	2.000	—	Edgerton (2011)
<i>Panel B: Parameters jointly calibrated to data moments</i>			
θ : Collateral multiplier	2.143	1.985	Debt-to-output ratio: 0.600
η_p : Productivity-tail parameter	5.949	5.397	Top-decile employment share: 0.632
ω : Productivity persistence	0.861	0.836	Firm exit rate: 0.098
σ_y : Ability innovation dispersion	0.245	0.239	Worker wage Gini: 0.372
β : Discount factor	0.858	0.854	Real interest rate: 0.031
ζ : New-to-used transition rate	0.079	0.000	New-to-used capital ratio: 2.570
κ : Breakage rate of used capital	0.143	—	Used-share gradient: -0.027

Notes: All targets are computed on the Vietnamese Small and Medium Enterprise Survey (CIEM et al., 2005–2015). The two economies are calibrated separately, as described below.

Comparison design. I compare two economies to isolate the role of the used-capital: a benchmark in which firms rent both new and used capital, and an economy without used capital. Both are calibrated to the same five common moments; the benchmark additionally matches the used-capital-share gradient and the aggregate new-to-used ratio. Removing used capital ($\zeta = 0$, $\phi = 1$) leaves the five common moments to be matched by recalibrating $\{\theta, \eta_p, \omega, \sigma_y, \beta\}$, and I solve for the general-equilibrium prices and allocations separately in each economy. Because both economies attain the same fit to the common targets, the differences between them reflect the absence of the used-capital. The lower value of θ is the value required to reproduce the benchmark’s debt-to-output ratio.

A. The Mechanism of Used Capital

Used capital works through the collateral constraint. A firm that cannot meet the constraint operates below its efficient scale. The multiplier μ measures the shadow cost of the missing financing capacity. The left panel of Figure 1 plots this shadow cost by benchmark entrepreneur wealth decile, separately for low, mid, and high entrepreneurial productivity z . It is highest for entrepreneurs who combine high productivity with little wealth—precisely the firms for which the constraint binds most strongly—and falls toward zero as wealth accumulates.

The right panel maps the shadow cost into the capital mix. An unconstrained firm faces $\mu = 0$ and chooses the mix of new and used capital that minimizes its rental cost

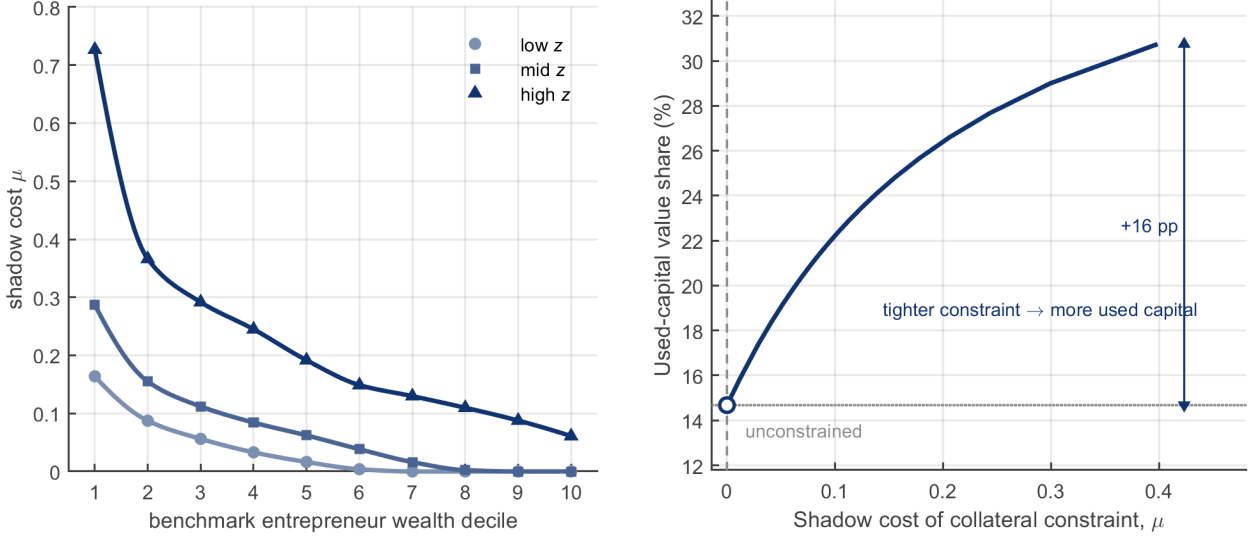


Figure 1. The mechanism of used capital. Left panel: the shadow cost of the collateral constraint, μ , by benchmark entrepreneur wealth decile, for low, mid, and high entrepreneurial productivity z . Right panel: the used-capital value share, $100 p_u K_u / (p_n K_n + p_u K_u)$, as a function of μ at benchmark equilibrium prices. The horizontal axis covers the range containing most constrained firms, and the open circle marks the unconstrained firm at $\mu = 0$.

at prevailing prices. Constrained firms depart from this benchmark. A dollar of rental spending on capital type j purchases $1/(p_j R_j)$ units, each of which absorbs p_j units of financing capacity. Consequently, a dollar of rental spending on type j uses $1/R_j$ units of financing capacity. Because $R_u > R_n$, used capital absorbs less financing capacity per dollar of rental spending than new capital. The used-capital share therefore rises monotonically with μ and approaches the ceiling implied by $\bar{\rho}$ in Proposition 2.

Taken together, the two panels show that the firms facing the tightest collateral constraints are precisely those that tilt most strongly toward used equipment. This tilt is a response to financing constraints rather than a difference in production technology. By reallocating financing capacity toward used equipment, constrained firms can obtain more effective capital from a given collateral position. This mechanism drives the allocation and entry results that follow.

B. Used Capital and Capital Misallocation

I measure the cost of financial frictions by comparing each calibrated economy with its own frictionless counterpart. In each case I set θ large enough that the collateral constraint is slack for every operating firm. I hold all other parameters fixed and recompute equilibrium prices and the stationary distribution. No agent with strictly positive wealth is then constrained, and MRPK dispersion is numerically zero. Using a separate frictionless counterpart for each economy isolates the misallocation caused by financial frictions from differences in technology, calibration, and equilibrium prices.

Define the TFP loss from the collateral constraint as

$$\text{TFP loss} = 100 \log \left(\frac{\text{TFP}^{\text{frictionless}}}{\text{TFP}^{\text{friction}}} \right), \quad (37)$$

measured in log points against the economy’s own frictionless counterpart.

Relative to its own frictionless counterpart, the collateral constraint generates a TFP loss of 5.72 log points in the benchmark. The same constraint generates 9.39 log points in the model without used capital, recalibrated to the same targets. Second-hand capital therefore offsets 3.67 log points, or

$$\frac{9.39 - 5.72}{9.39} = 39.1\%$$

of the frictional loss that arises in its absence. This decomposes the model’s misallocation into the portion the used-capital margin removes. Table 4 reports this comparison together with the allocation statistics behind it.

Table 4. Used Capital and Capital Misallocation

	Benchmark	No used capital	Effect of used capital
TFP loss (log points)	5.72	9.39	39.1% eliminated
Mean scale relative to target, K/K^*	0.343	0.279	23.0% higher
sd(log MRPK)	0.450	0.598	24.7% lower
Share of firms constrained	0.908	0.909	0.2% lower

Notes: The benchmark contains new and used capital, while the other model is without used capital and recalibrated to the common targets. The TFP loss is defined in (37) and is computed within each economy against its own frictionless counterpart, so the two columns are not measured against a common benchmark. Percentages in the last column compare the benchmark with the model without used capital and are computed from the unrounded values.

Second-hand capital reallocates capital toward financially constrained, high-return firms. Access to used capital allows constrained entrepreneurs to operate closer to their unconstrained scale, as Figure 2 shows. The effect is largest among low-wealth entrepreneurs. Second-hand capital therefore directs financing capacity toward firms with the highest marginal returns to capital.

The single-vintage economy is separately calibrated. The headline estimate therefore also incorporates changes in the fitted productivity distribution. Holding that distribution fixed instead gives 19.5 percent. Appendix C reports all three comparison designs. No margin changes sign across them.

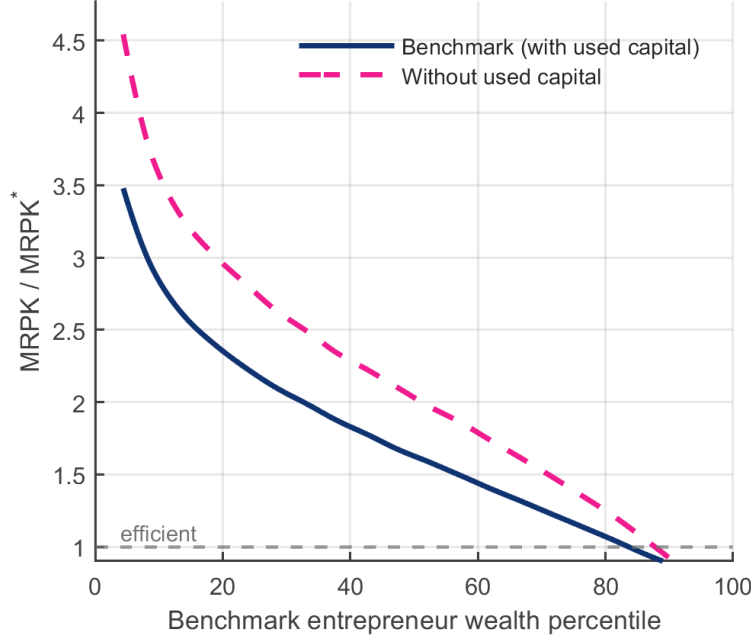


Figure 2. Used capital and the allocation of capital. MRPK relative to each economy’s own frictionless level, by benchmark entrepreneur wealth percentile, in the benchmark (solid) and the model without used capital (dashed).

C. Used Capital and Entry by Productive Poor Agents

The second main result concerns occupational selection. I define “talented” agents as those in the top decile of entrepreneurial productivity z . “Poor” agents are those with wealth below the benchmark median. Access to used capital raises entrepreneurship among poor talented agents from 34.3 to 42.0 percent, an increase of 7.7 percentage points. By contrast, entrepreneurship among rich talented agents is nearly unchanged. Used capital therefore reduces the dependence of entrepreneurship on initial wealth rather than increasing entry uniformly. Table 5 reports these entry rates together with their frictionless counterparts.

Table 5. Used Capital and Entry by Productive Poor Agents

	Benchmark	No used capital	Effect of used capital
Poor talented entrepreneurship rate	0.420	0.343	+7.7 p.p.
Rich talented entrepreneurship rate	0.827	0.833	−0.6 p.p.
Poor talented rate, frictionless	0.714	0.682	—
Rich talented rate, frictionless	0.801	0.744	—
Entry shortfall, $\mathcal{G}$ (log points)	53.2	68.7	22.6% repaired

Notes: “Talented” denotes the top productivity decile and “poor” denotes wealth below the benchmark population median. These cutoffs are constructed once in the benchmark and applied in levels to both economies. Each economy is compared with its own frictionless counterpart, so the frictionless rows differ across columns. The entry shortfall $\mathcal{G}$ in (38) is defined for poor talented agents.

The mechanism operates through the occupational boundary. Used capital absorbs less financing capacity. Productive agents with little wealth can therefore operate at a larger scale. This raises profits $\pi(a, z)$ and the value of entrepreneurship $V_e(a, z, y)$, enabling some of them to enter. Rich talented agents already possess sufficient financing capacity. Their occupational choices therefore change little.

To account for poor talented agents who prefer wage work even without financial frictions, I define the collateral-induced entry shortfall as

$$\mathcal{G} = 100 \log \left(\frac{e^{\text{frictionless}}}{e^{\text{friction}}} \right), \quad (38)$$

where e is their entrepreneurship rate. The shortfall is 53.2 log points in the benchmark and 68.7 log points in the model without used capital. Thus, used capital repairs

$$1 - \frac{53.2}{68.7} = 22.6\%$$

of the entry distortion caused by the collateral constraint.

For rich talented agents, the shortfall is negative in both economies. Entry under the constraint (0.827 in the benchmark) exceeds entry in its own frictionless counterpart (0.801), implying a shortfall of -3.2 log points. The corresponding figure without used capital is -11.3 log points. This is a general-equilibrium effect rather than a relaxation of the constraint. Removing the collateral constraint raises labor demand and hence the wage. This raises the value of wage work and draws some marginal high-productivity agents out of entrepreneurship. The effect is confined to agents whose financing capacity was never binding.

IV. Conclusion

This paper begins from a discrepancy between standard models and the data. In models with homogeneous capital, a financially constrained firm can respond to limited credit only by operating at a smaller scale. In the data, however, young, small, and credit-excluded firms also change the mix of their capital. They rely more heavily on second-hand equipment. This paper studies whether that additional adjustment margin changes the allocation and occupational consequences of financial frictions. The mechanism follows from the way collateral constraints are defined. The constraint limits the market value of the capital a firm operates. Used capital is more expensive to operate, because it depreciates faster. But its market price is lower. A dollar spent renting used equipment therefore ties up less collateral than a dollar spent renting new. A productive entrepreneur with little wealth can use second-hand equipment

to operate more capital than new equipment alone would allow. As the entrepreneur accumulates wealth, the constraint becomes less severe. The advantage then declines, and the firm shifts toward the unconstrained mix. This mechanism accounts for the concentration of used equipment among young, asset-poor, and credit-excluded firms in the Vietnamese data. Quantitatively, access to second-hand capital reduces the misallocation that the collateral constraint generates by 39 percent. The comparison is with a single-vintage economy in which firms rent only new capital. It is calibrated on its own to the same five empirical targets; because the vintage margin is absent, its calibrated parameters differ from the benchmark's. Used equipment allows productive firms with little wealth to operate closer to their efficient scale, and it reduces the dispersion of marginal products across firms. The same mechanism affects occupational choice. Without access to used capital, productive but poor agents are less likely to become entrepreneurs. Entry among productive and wealthy agents changes little.

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Appendix A. The Intermediary’s Problem

Proof of Proposition 1. The dynamic problem of the intermediary is

$$V(K_n, K_u) = \max_{K'_n, I} \left\{ R_n K_n + p_u R_u K_u - I + \frac{V(K'_n, K'_u)}{1+r} \right\}, \quad (\text{A1})$$

subject to

$$K'_n = (1 - \delta - \zeta)K_n + I, \quad K'_u = (1 - \delta - \kappa)K_u + \zeta K_n.$$

Take the first-order derivatives with respect to K_n and K_u :

$$\frac{\partial V(K_n, K_u)}{\partial K_n} = R_n + (1 - \delta - \zeta) + \frac{\partial V(K'_n, K'_u)}{\partial K'_n} \frac{\zeta}{1+r}, \quad (\text{A2})$$

$$\frac{\partial V(K_n, K_u)}{\partial K_u} = p_u R_u + \frac{\partial V(K'_n, K'_u)}{\partial K'_u} \frac{1 - \delta - \kappa}{1+r}. \quad (\text{A3})$$

The first-order condition with respect to K'_n is

$$\frac{\partial V(K_n, K_u)}{\partial K'_n} = -1 + \frac{\partial V(K'_n, K'_u)}{\partial K'_n} \frac{1}{1+r} = 0. \quad (\text{A4})$$

Equation (A3) implies

$$\frac{\partial V(K_n, K_u)}{\partial K_u} = p_u R_u + \frac{1 - \delta - \kappa}{1+r} p'_u R'_u + \left(\frac{1 - \delta - \kappa}{1+r} \right)^2 p''_u R''_u + \dots$$

At the steady state this is equivalent to

$$\frac{\partial V(K_n, K_u)}{\partial K_u} = p_u R_u \frac{1+r}{r+\delta+\kappa}. \quad (\text{A5})$$

Combine (A2) and (A4) to obtain

$$1+r = R_n + (1-\delta-\zeta) + \frac{\zeta}{1+r} \frac{\partial V(K_n'', K_u'')}{\partial K_u''}. \quad (\text{A6})$$

Applying the steady-state condition (A5),

$$r = R_n - \delta - \zeta \left(1 - \frac{p_u R_u}{r+\delta+\kappa} \right), \quad (\text{A7})$$

which is (18). □

The split into a price and a user cost. Only the product $p_u R_u$ appears in (A7). Define $R_u = r + \delta + \kappa$. Then (A5) becomes $\partial V / \partial K_u = (1+r)p_u$, so p_u is the value of a used unit net of its current rental. Equation (A7) now reads $R_n = r + \delta + \zeta(p_n - p_u)$, and inverting it delivers (20). Only the product enters any firm's rental bill. The financing constraint (9), written on market value, is what makes the two factors matter separately.

Zero profit. The technology is constant returns and (A1) is linear, so V is homogeneous of degree one. By Euler's theorem, $V = V_n K_n^s + V_u K_u^s = (1+r)(p_n K_n^s + p_u K_u^s)$. Deposits are the only liability and capital the only asset, which is (21). The intermediary therefore owes $(1+r)D$ against a portfolio worth V , and profit is zero. This establishes the claim made in Section II.D.

Appendix B. Derivations

A. The capital mix

Proof of Proposition 2(i). A physical unit of type j costs $p_j R_j$ to rent and absorbs p_j of financing capacity. At shadow cost $\mu \geq 0$ its effective price is therefore $p_j(R_j + \mu)$, and for a given level K of effective capital the entrepreneur solves

$$\min_{K_n, K_u} p_n(R_n + \mu)K_n + p_u(R_u + \mu)K_u \quad \text{s.t.} \quad \left[\phi^{1/\gamma} K_n^{\frac{\gamma-1}{\gamma}} + (1-\phi)^{1/\gamma} K_u^{\frac{\gamma-1}{\gamma}} \right]^{\frac{\gamma}{\gamma-1}} = K. \quad (\text{B1})$$

The first-order conditions give the CES demand ratio (31), which factors as

$$\rho(\mu) = \bar{\rho} \left[\frac{R_n + \mu}{R_u + \mu} \right]^\gamma, \quad \bar{\rho} \equiv \frac{1 - \phi}{\phi} \left(\frac{p_n}{p_u} \right)^\gamma, \quad (\text{B2})$$

since the price ratio does not depend on μ . In particular $\rho^{uc} = \bar{\rho} (R_n/R_u)^\gamma$. Differentiating the bracket,

$$\frac{\partial}{\partial \mu} \frac{R_n + \mu}{R_u + \mu} = \frac{R_u - R_n}{(R_u + \mu)^2}, \quad (\text{B3})$$

so $\rho'(\mu) > 0$ if and only if $R_u > R_n$, which is (33). As $\mu \rightarrow \infty$ the bracket tends to one. Hence $\rho(\mu) \rightarrow \bar{\rho}$. $\square$

B. Collateral productivity

Proof of Proposition 2(ii). Normalize $K_n = 1$, so that $\rho = K_u$ and the bundle is $\Psi(\rho) = [\phi^{1/\gamma} + (1 - \phi)^{1/\gamma} \rho^{\frac{\gamma-1}{\gamma}}]^\frac{\gamma}{\gamma-1}$. The capacity absorbed is $p_n + p_u \rho$, so collateral productivity is $\Lambda(\rho) = \Psi(\rho)/(p_n + p_u \rho)$, as in (32). Maximizing Λ is equivalent to minimizing the collateral cost of one unit of the bundle, $(p_n + p_u \rho)/\Psi(\rho)$. That function is strictly quasi-convex in ρ : for every $t > 0$ the map $\rho \mapsto p_n + p_u \rho - t\Psi(\rho)$ is strictly convex, since Ψ is strictly concave, so each sublevel set is an interval. Its minimizer is therefore unique, and the first-order condition returns the $\bar{\rho}$ of (B2): the mix that minimizes collateral at prices (p_n, p_u) is the same mix that (B2) approaches as $\mu \rightarrow \infty$, because the user costs drop out of the demand ratio in that limit. Hence

$$\bar{\Lambda} \equiv \Lambda(\bar{\rho}) = \left[\phi p_n^{1-\gamma} + (1 - \phi) p_u^{1-\gamma} \right]^{1/(\gamma-1)}, \quad (\text{B4})$$

which is (34), and Λ is strictly increasing on $(0, \bar{\rho})$.

By (B2), $\rho(\mu)$ is $\bar{\rho}$ times a bracket that under (33) rises to one from below. Therefore $\rho(\mu)$ increases from $\rho^{uc} < \bar{\rho}$ toward $\bar{\rho}$, and $\Lambda(\rho(\mu))$ is strictly increasing in μ with limit $\bar{\Lambda}$. Finally $\bar{\Lambda} > p_n^{-1}$ whenever $p_u < p_n$ and $\gamma > 1$, since then $p_u^{1-\gamma} > p_n^{1-\gamma}$ and the convex combination in (B4) exceeds $p_n^{1-\gamma}$. $\square$

Two benchmarks should be distinguished. The p_n^{-1} of Proposition 2(ii) is collateral productivity in the economy without used capital, where $K = K_n$ and a dollar of collateral supports $1/p_n$ units of the bundle; it is the counterpart of the recalibrated no-used economy of Section III. It is not $\Lambda(0) = \phi^{1/(\gamma-1)}/p_n$, the productivity of a firm that retains the share parameter $\phi < 1$ but is denied used capital. Since $\phi < 1$ and $\gamma > 1$, $\Lambda(0) < p_n^{-1}$, so the ratio $\bar{\Lambda}/\Lambda(0)$ combines the financing gain with a mechanical penalty from a missing CES input; $\bar{\Lambda} > \Lambda(0)$ holds by optimality alone and requires

no condition on prices.

C. Which margin the relief runs on

Three price comparisons should be kept apart. Used capital is cheaper per physical unit, $p_u < p_n$, which by (20) is equivalent to $R_n > r + \delta$; this is what delivers $\bar{\Lambda} > p_n^{-1}$, and prices are the whole of that result. Its user cost per dollar of market value is higher, $R_u > R_n$ under (33); this is what delivers the tilt. Its rental payment per physical unit, $p_u R_u$, may lie on either side of $p_n R_n$, and no result here depends on which. Both $p_u < p_n$ and $R_u > R_n$ are equilibrium properties rather than restrictions on primitives, and both hold at the calibrated values.

The effective price $p_j(R_j + \mu)$ in (B1) already encodes the trade-off, and reading it per rental dollar makes the mechanism plain. A dollar of rental spending on type j hires $1/(p_j R_j)$ units and ties up

$$\frac{p_j}{p_j R_j} = \frac{1}{R_j} \quad (\text{B5})$$

of financing capacity. The price cancels, so on this margin the user cost alone decides: renting used capital frees collateral relative to new exactly when $R_u > R_n$, the equivalent forms of which are given in Proposition 2. What the constraint counts is market value, and a rental dollar spent on used capital commands less of it.

Appendix C. The Comparison Design and the Productivity Tail

The comparison in the text recalibrates the economy without used capital in full. Its Pareto-tail parameter therefore falls from 5.949 to 5.397. A fatter tail places more mass on highly productive entrepreneurs, who have the largest desired scale, and so raises measured misallocation on its own. This appendix holds the tail fixed in two ways. The *tail-pinned* economy keeps $(\eta_p, \omega, \sigma_y)$ at their benchmark values and recalibrates only (θ, β) . The *tail diagnostic* returns η_p to 5.949 in the fully recalibrated economy and re-clears prices without refitting any moment. Its targeted moments therefore no longer hold exactly. Misallocation averted is one minus the ratio of the benchmark's TFP loss to that design's TFP loss, with the TFP loss defined in (37). Entry repaired is defined the same way from the entry shortfall in (38). Each economy is compared with its own frictionless twin. Each economy is compared with its own frictionless twin. Changing η_p alone moves the TFP loss from 7.42 to 9.39 log points. The collateral multiplier, productivity persistence, worker dispersion, the discount factor and equilibrium prices together account for 0.31. Under all three designs the benchmark has the lower TFP loss, the higher mean scale relative to target, the lower dispersion of log MRPK, and the higher rate of entry by poor talented agents. The

Table 6. The result under three designs without used capital

Economy	η_p	TFP loss (log pts)	Misallocation averted	Entry repaired	K/K^*	sd(logMRPK)
Benchmark (used K)	5.949	5.72	—	—	0.343	0.450
No used capital, tail pinned	5.949	7.11	19.5%	32.0%	0.268	0.530
Tail diagnostic	5.949	7.42	22.9%	27.7%	0.295	0.530
No used capital, recalibrated	5.397	9.39	39.1%	22.6%	0.279	0.598

share of the frictional loss attributed to used capital varies with the design. Its sign does not. The two margins move in opposite directions. Misallocation averted rises from 19.5 to 39.1 percent as the tail is refit, while entry repaired falls from 32.0 to 22.6. A fatter tail places more mass on very productive entrepreneurs, whose desired scale is largest, so constraining them is expensive and measured misallocation rises. But those same agents are inframarginal in the occupational choice—productive enough to enter even at a constrained scale—so the constraint blocks fewer of them and the entry shortfall is smaller.

Appendix D. Robustness to the Assigned CES Parameters

The weight ϕ on new capital and the elasticity of substitution γ in the capital aggregator (4) are assigned rather than calibrated; the benchmark sets $\phi = 0.70$ and $\gamma = 2.0$. Table 7 re-solves the benchmark at $\phi = 0.75$, the value used by Lanteri (2018), and at $\gamma \in \{1.7, 2.5\}$, which brackets the range Edgerton (2011) estimates, in each case holding the calibrated parameters $(\theta, \eta_p, \omega, \kappa, \sigma_y, \beta)$ at their benchmark values and re-clearing only the prices (W, r, R_n) . These are robustness sweeps rather than recalibrations: each asks whether the results turn on the assigned value, not whether the model could be re-fitted around it. The two sweeps were solved in separate runs and should each be read against their own reference. Neither parameter matters for the conclusions: the misallocation loss moves by at most 0.06 log points, about one percent of its level, and misallocation averted stays between 38.7 and 39.3 percent.

Table 7. Sensitivity of the results to the assigned CES parameters

	ϕ sweep		γ sweep	
	$\phi = 0.70$ (benchmark)	$\phi = 0.75$	$\gamma = 1.7$	$\gamma = 2.5$
<i>Panel A: Gain from removing the collateral constraint</i>				
TFP (%)	5.89	5.91	5.92	5.86
Misallocation loss $\mathcal{L}$ (log points)	5.72	5.74	5.75	5.70
Misallocation averted (%)	39.1	38.9	38.7	39.3
<i>Panel B: Allocation in the constrained equilibrium</i>				
Mean scale relative to target, K/K^*	0.343	0.350	0.347	0.337
Share of firms constrained	0.908	0.906	0.907	0.908
Entrepreneur share	0.221	0.220	0.221	0.220
<i>Panel C: Cleared prices</i>				
Wage W	0.540	0.542	0.540	0.540
Interest rate r	0.031	0.031	0.031	0.031
New-capital rental R_n	0.119	0.124	0.119	0.120
Used-capital price p_u	0.645	0.582	0.648	0.640

Notes: Each column re-clears (W, r, R_n) at the stated parameter value with the calibrated parameters held at their benchmark levels; the benchmark is $\phi = 0.70$, $\gamma = 2.0$. $\mathcal{L}$ is the TFP loss from the collateral constraint defined in (37), measured against each economy's own frictionless counterpart; the TFP row is the same object in percent rather than log points.